

Deutsch Algorithm

①

We are presented with a black box function (Oracle) that receives a single qubit as input and produces corresponding o/p in one of two ways. Either the function always returns same value

$$\begin{array}{l} f(0) = 0 = f(1) \\ f(1) = 1 = f(0) \end{array}$$

Category I $\rightarrow$ Constant

OR

$$\begin{array}{ll} f(0) = 1 & f(1) = 0 \\ f(0) = 0 & f(1) = 1 \end{array}$$

Category 2 $\rightarrow$ balanced

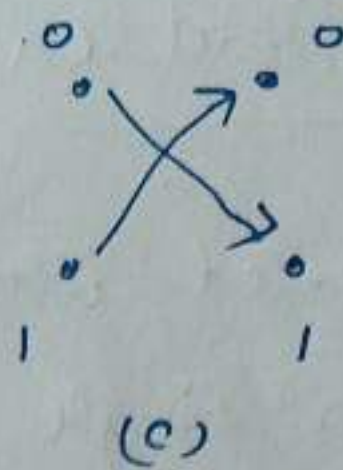
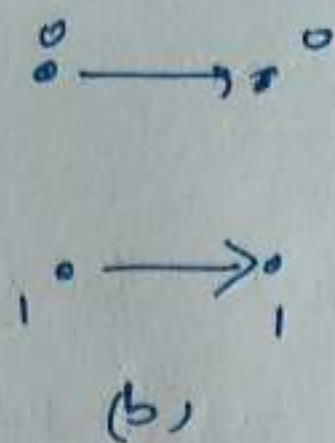
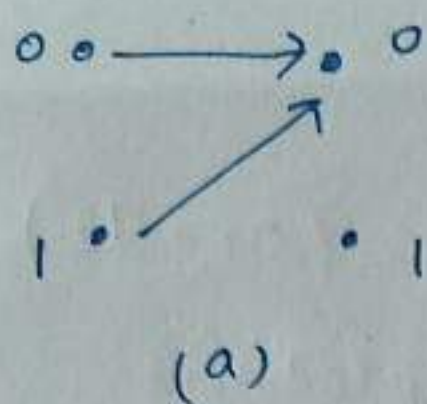
The objective is to determine if the function is balanced or constant.

Classical bits:- Test for both inputs $f(0)$ & $f(1)$ then find if function returns same value or diff. value.

If same Then constant
If different Then balanced.

With classical computer we need to evaluate f twice.
Can we do better using quantum computer?

Deutsch Algo is concerned with functions from the set $\{0,1\}$ to the set $\{0,1\}$. The four such functions that can be visualized are:-

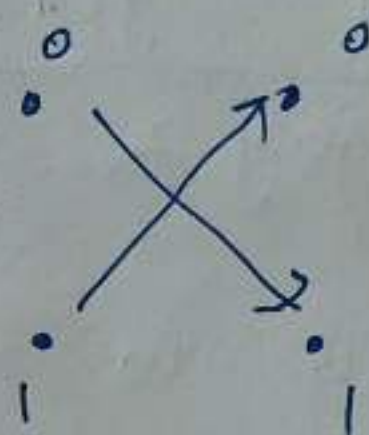


(a) & (d) $\rightarrow$ ~~constant~~ constant

(b) & (c) $\rightarrow$ balanced.

Deutsch Algo:-

consider the f^n :-



$$f(0) = 1$$

$$f(1) = 0$$

lets find a Matrix such that

$$M|0\rangle = |1\rangle$$

$$M|1\rangle = |0\rangle$$

columns of Matrix are output of basis inputs.

$|0\rangle$ gives $|1\rangle$ $\therefore$ 1st colⁿ is $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$|1\rangle$ gives $|0\rangle$ $\therefore$ 2nd colⁿ is $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$

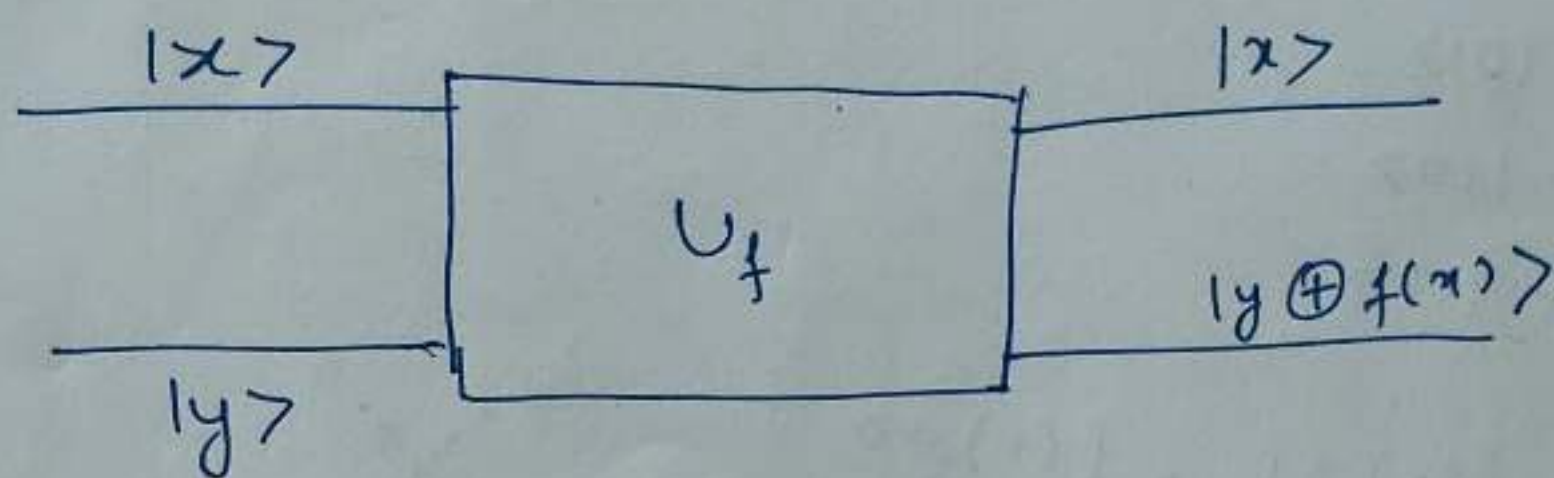
$$\therefore M = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

lets verify $M|0\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle$

$$M|1\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle$$

Describe the matrices for the other three functions $\{0,1\}$ to $\{0,1\}$.

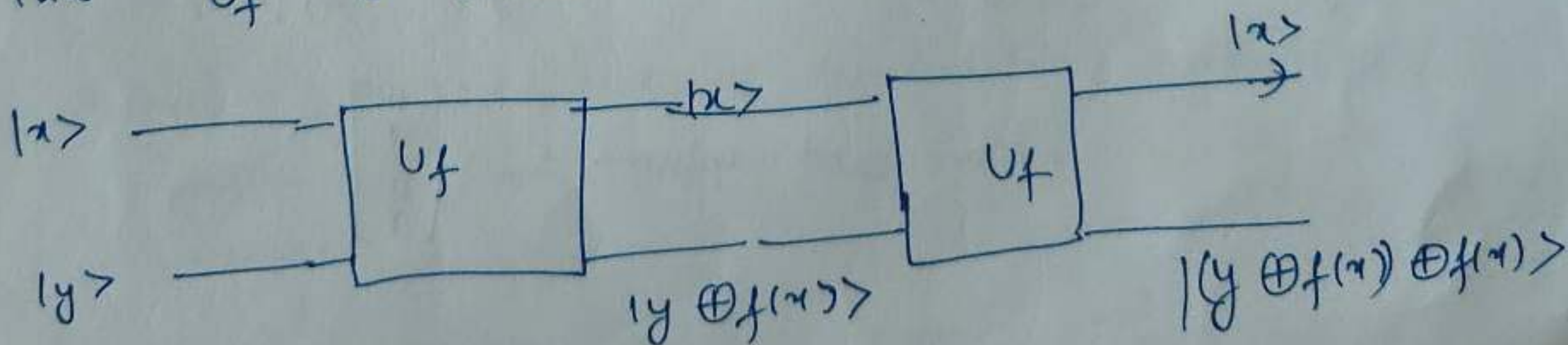
Let's
Define a gate for quantum system. The gate must be unitary (& thus reversible).



The top qubit will be the qubit which will be evaluated & bottom qubit is the control qubit.
The oracle keeps $|x>$ same & changes $|y>$ as $|y \oplus f(x)>$ where $\oplus$ is XOR.

$\therefore$ If $y=0$, $|x, 0> \rightarrow |x, 0 \oplus f(x)>$
If $y=1$, $|x, 1> \rightarrow |x, 1 \oplus f(x)>$

This U_f is reversible



New $|x, y> \rightarrow |x, (y \oplus f(x)) \oplus f(x)>$

$$|x, y\rangle \rightarrow |x, y \oplus (f(x) \oplus f(x))\rangle \rightarrow |x, y \oplus 0\rangle$$

lets find U_f .

for two qubits the basis states are $|00\rangle, |01\rangle, |10\rangle, |11\rangle$

These are four columns of matrix

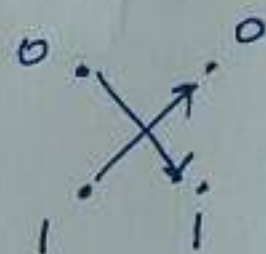
column 1: Input $|00\rangle$

column 2: input $|01\rangle$

column 3: input $|10\rangle$

column 4: Input $|11\rangle$

for function $f(0)=1, f(1)=0$



① Input $|00\rangle, x=0, y=0$

Rule $|x, y\rangle \rightarrow |x, y \oplus f(x)\rangle$

$$|0, 0\rangle \rightarrow |0, 0 \oplus f(0)\rangle \rightarrow |0, 0 \oplus 1\rangle \rightarrow |01\rangle$$

$\therefore$ 1st column is

$$\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

② Input $|01\rangle, x=0, y=1$

$$|x, y\rangle \rightarrow |x, y \oplus f(x)\rangle \rightarrow |0, 1 \oplus f(0)\rangle \rightarrow |0, 1 \oplus 1\rangle \rightarrow |00\rangle$$

$\therefore$ 2nd column is

$$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Input $|10\rangle$ $x=1, y=0$

$$|xy\rangle \Rightarrow |x, y \oplus f(x)\rangle \rightarrow |1, 0 \oplus f(1)\rangle = |1, 0 \oplus 0\rangle = |10\rangle$$

3rd colⁿ is $\begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$

Input $|11\rangle$ $x=1, y=1$

$$|xy\rangle = |1, 1 \oplus f(1)\rangle = |1, 1 \oplus 0\rangle = |11\rangle$$

4th colⁿ is $\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$

$$\therefore U_f = \begin{matrix} & \begin{matrix} 00 & 01 & 10 & 11 \end{matrix} \\ \begin{matrix} 00 \\ 01 \\ 10 \\ 11 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix} \quad \text{--- (1)}$$

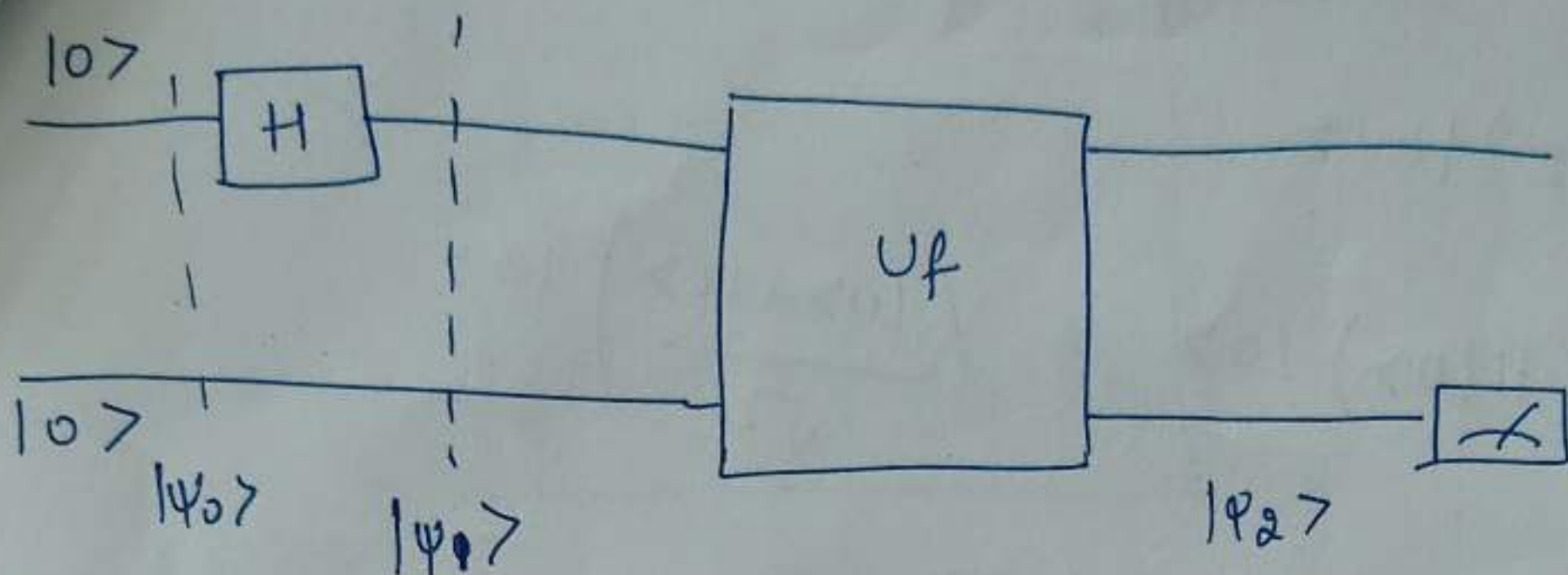
Here Top column name represents to input $|x, y\rangle$ and the left-hand row name corresponds to the outputs $|x', y'\rangle$.
A '1' in a ' xy ' colⁿ and ' $x'y'$ ' row means that for input $|x, y\rangle$, the output will be $|x', y'\rangle$

Que 2 What is the adjoint of the matrix given in equation 1. Show that this matrix is its own

Variation 1

(11)

(6)



Let input state be $|0\rangle$ or $|1\rangle$, put top input state to $\frac{|0\rangle + |1\rangle}{\sqrt{2}}$ (half way $|0\rangle$ & half way $|1\rangle$)

we can use hadamard gate to achieve it

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

The output of ckt corresponds to

$$U_f (H \otimes I) (|00\rangle) \text{ or } U_f (H \otimes I) (|0\rangle \otimes |0\rangle) \rightarrow \textcircled{X}$$

Now $|00\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$

$\therefore \textcircled{X}$ becomes

$$U_f (H \otimes I) \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\psi_0 = |0\rangle \otimes |0\rangle = |00\rangle$$

$$\psi_1 = (H \otimes I) |00\rangle$$

$$= (H|0\rangle) |0\rangle = \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) |0\rangle$$

$$= \frac{|00\rangle + |10\rangle}{\sqrt{2}}$$

$$\psi_2 = U_f(\psi_1) \quad \text{-----} \quad (*)$$

$$= \frac{|0, 0 \otimes f(0)\rangle + |1, 0 \otimes f(1)\rangle}{\sqrt{2}}$$

$$\text{Now } f(0)=1, f(1)=0$$

$$\text{1st term } |0, 0 \otimes f(0)\rangle = |0, 0 \otimes 1\rangle = |0, 1\rangle$$

$$\text{2nd term } |1, 0 \otimes f(1)\rangle = |1, 0 \otimes 0\rangle = |1, 0\rangle$$

$$\therefore \psi_2 = \frac{|0, 1\rangle + |1, 0\rangle}{\sqrt{2}}$$

On measurement, 50-50 chances that input qubit is $|0\rangle$ or $|1\rangle$

implies no real information gotten by measuring bottom qubit.

Ques 11 Use the matrix U_f to find ψ_2 (in $(*)$)

See the matrix U_f , and determine the state
2 for the other three functions

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$$f(0) = 0 = f(1)$$

$$f(0) = 1 = f(1)$$

$$f(0) = 0, f(1) = 1$$

0

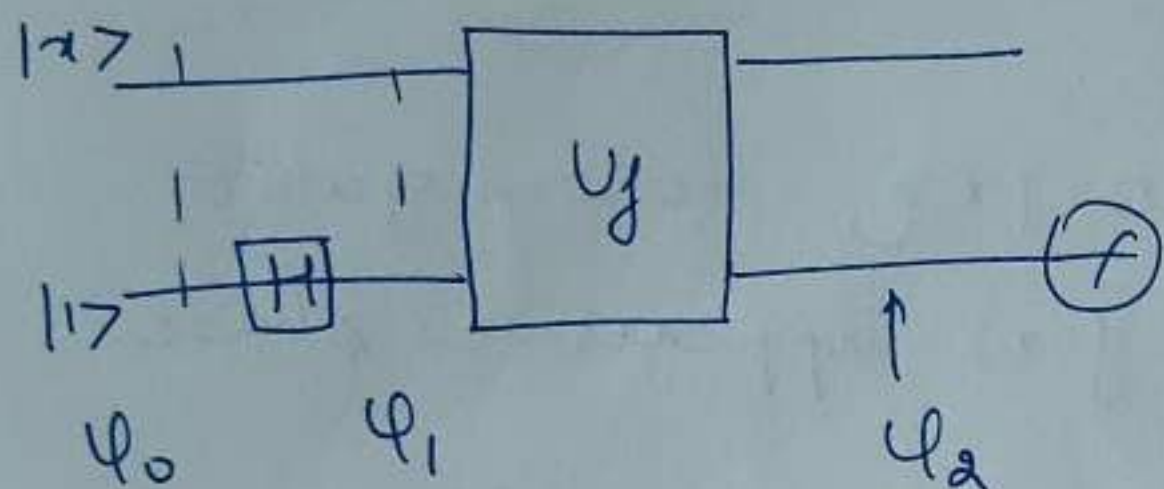
0

0

(9)

Variation 2

Let's put bottom qubit to $H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$ superposition state and keep input qubit in state $|x\rangle$ where $|x\rangle = |0\rangle$ or $|x\rangle = |1\rangle$



Circuit is $U_f (I \otimes H) (|x, 1\rangle)$

$$\psi_0 = |x, 1\rangle$$

$$\psi_1 = (I \otimes H) (|x, 1\rangle)$$

$$= |x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) = \frac{|x, 0\rangle - |x, 1\rangle}{\sqrt{2}}$$

$$\psi_2 = U_f \left(\frac{|x, 0\rangle - |x, 1\rangle}{\sqrt{2}} \right)$$

$$= \frac{|x, 0 \oplus f(x)\rangle - |x, 1 \oplus f(x)\rangle}{\sqrt{2}}$$

When $f(x) = 0$

$$\psi_2 = \frac{|x, 0 \oplus 0\rangle - |x, 1 \oplus 0\rangle}{\sqrt{2}} = \frac{|x, 0\rangle - |x, 1\rangle}{\sqrt{2}}$$

$$= \frac{|x\rangle (|0\rangle - |1\rangle)}{\sqrt{2}}$$

$$\boxed{\text{when } f(x)=1}$$

$$\psi_2 = \frac{|x, 0 \oplus 1\rangle - |x, 1 \oplus 1\rangle}{\sqrt{2}} = \frac{|x, 1\rangle - |x, 0\rangle}{\sqrt{2}}$$

$$= -|x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

Top qubit will be in state $|x\rangle$, bottom qubit will be either $|0\rangle$ or $|1\rangle$. $f(x)$ appears as phase.

$$\psi_2 = \begin{cases} |x\rangle \frac{|0\rangle - |1\rangle}{\sqrt{2}}, & f(x)=0 \\ |x\rangle \frac{|1\rangle - |0\rangle}{\sqrt{2}}, & f(x)=1 \end{cases}$$

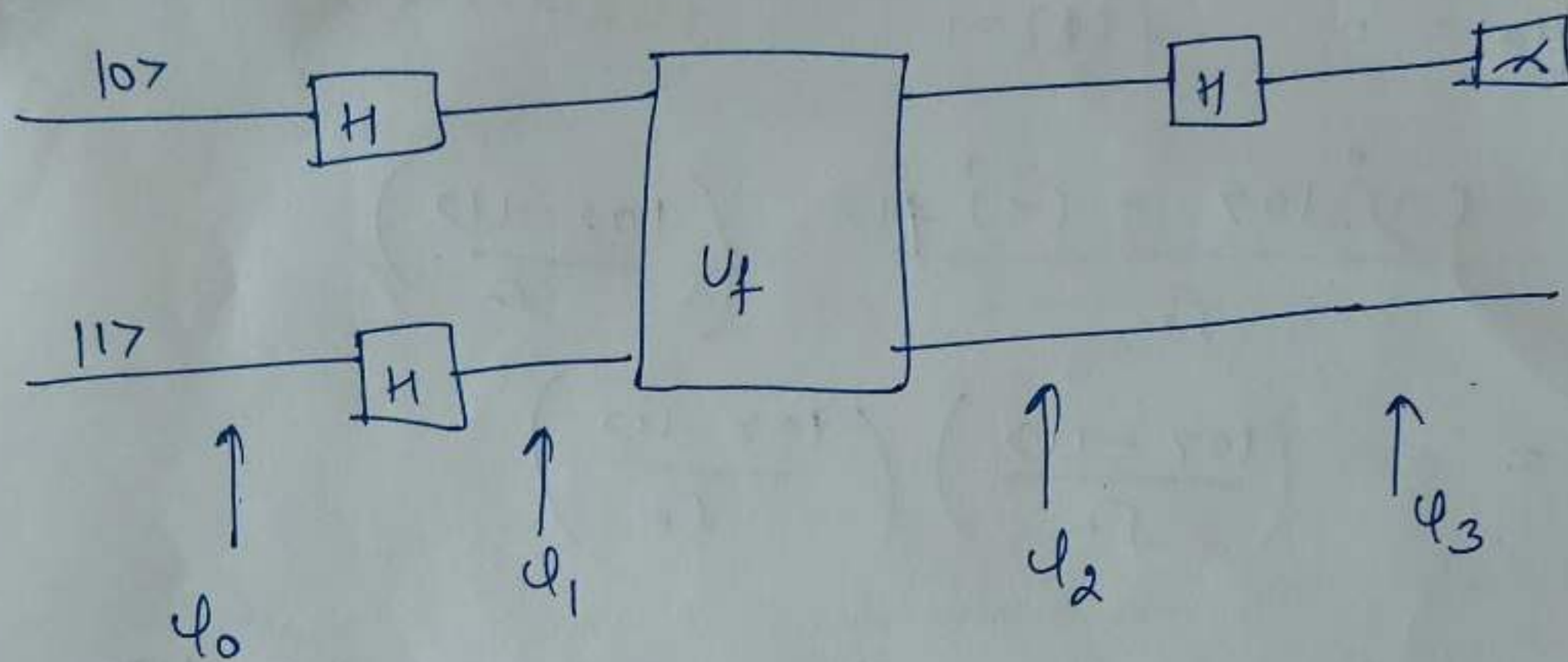
$$\text{or } \boxed{\psi_2 = (-1)^{f(x)} |x\rangle \left[\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right]}$$

when bottom qubit is $|1\rangle$, $f(x)$ appears as
phase
=

Deutsch Algorithm

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Put both top & bottom qubits into a superposition.



$$(H \otimes I) U_f (H \otimes H) |0, 1\rangle$$

$$\phi_1 = (H \otimes H) |0, 1\rangle$$

$$= \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

$$\frac{|00\rangle - |01\rangle + |10\rangle - |11\rangle}{2}$$

$$|\phi_2\rangle = (-1)^{f(x)} |x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

when $x = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$

↓

$$|\phi_2\rangle = \frac{(-1)^{f(0)} |0\rangle + (-1)^{f(1)} |1\rangle}{\sqrt{2}} \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

$$|\varphi_2\rangle = \frac{(-1)^{f(0)}|0\rangle + (-1)^{f(1)}|1\rangle}{\sqrt{2}} \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

$$f(0) = 0 \quad f(1) = 1 \quad (\text{balanced})$$

$$\psi_2 = \frac{(-1)^0 |07\rangle + (-1)^1 |11\rangle}{\sqrt{2}} = \frac{|07\rangle - |11\rangle}{\sqrt{2}}$$

$$= \left(\frac{107 - 117}{\sqrt{2}} \right) \left(\frac{107 - 117}{\sqrt{2}} \right)$$

Measure r_1 Qubit H $(107-112)$ r_2 $(107+112)$

A hand-drawn graph of a periodic function, likely a sine wave, plotted on a grid. The function oscillates between approximately -1 and 1. The graph shows several cycles, with labels '1' and '-1' indicating the amplitude. The function is drawn in blue ink.

$$f(0) = 1 \quad f(\psi) = 1 \quad (\text{constant})$$

$$|\psi_2\rangle = \left(\frac{-|0\rangle + |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

$$= (-1) \left(\frac{107 + 117}{\sqrt{2}} \right) \left(\frac{107 - 117}{\sqrt{2}} \right)$$

$$f(0) = 0 \quad f(1) = 0$$

$$|p_2\rangle = \left(\frac{107+117}{\sqrt{2}} \right) \left(\frac{107-117}{\sqrt{2}} \right) \quad (\text{constant})$$

$f(0)=1$ $f(1)=0$ (balanced)

$$1425 = - \left(\frac{107 + 11\sqrt{5}}{\sqrt{2}} \right) \left(\frac{107 - 11\sqrt{5}}{\sqrt{2}} \right)$$

when f is balanced, it becomes

(13)

$$\pm \left(\frac{107 - 11}{\sqrt{2}} \right) \left(\frac{107 - 11}{\sqrt{2}} \right)$$

when f is constant, it becomes

$$\pm \left(\frac{107 + 11}{\sqrt{2}} \right) \left(\frac{107 - 11}{\sqrt{2}} \right)$$

Now Apply Hadamard on 1st Qubit.
Hadamard matrix is its own inverse

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$\therefore$ when f is constant, ~~it is~~ ^{0|1 is}
becomes $\pm 107 \left(\frac{107 - 11}{\sqrt{2}} \right)$

when f is balanced, 0|1 is

$$\pm 11 \left(\frac{107 - 11}{\sqrt{2}} \right)$$

measure, if you get 107, f is constant.
" " 11, f is balanced.